



UNIVERSITY OF
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PUSL3190 Computing Project

Abstract

An Online Platform for Saltern Staff
and Buyers

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Abstract

The traditional salt industry is prominent in Sri Lanka; however, many of the tasks related to its operations, such as monitoring of production, inventory management, and handling customer transactions, continue to be handled manually. This results in duplication of information, poor real-time visibility, delayed reporting, and inefficiency. Moreover, the customers do not have access to a platform that allows them to inquire about product availability, process their orders, pay for the products, and monitor the delivery process. The overall objective of the proposed study is to design and develop a scalable and secure web-based platform that integrates all aspects of the operations of the salterns and customer services. The unique aspect of the system being designed is that it will cater to the special requirements of salterns, such as brine concentration calculation, production according to weather, and yield prediction, which cannot be achieved by the ERP systems currently available on the market. The methodology for developing the proposed system includes document analysis, observations, and informal interviews with saltern employees. The proposed system was developed using MERN stack. Communication within the system was using RESTful APIs, while the JWT approach will ensure role-based authentication and security. Testing was done through functional tests, usability evaluation, and performance testing to gauge the effectiveness of the proposed platform. Preliminary evaluation reveals that the system provides better operational efficiency, reduces manual mistakes, improves reporting speed, and engages customers more effectively.